Zoë Y. Valladares

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Education

Duke University, Pratt School of Engineering — Durham, NC
Master of Engineering, Design & Technology Innovation — *Spring 2027*

Relevant Coursework: Design Climate I & II Fellowship, Applying Machine Learning to Advance Cybersecurity, Continuum Robotics

University of Delaware, College of Engineering — Newark, DE
Bachelor in Computer Science — *May 2024*

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Web App Security, Cloud Cryptography, Game Development

Experience

Markham LLC — Washington, DC
API, Automation, & AI Integration — *Sep 2024 - Present*

- Took initiative as an intern to eliminate manual workflows by developing an RPA solution with Raspberry Pi, Robomotion, and Quickbase Pipelines — automating multiple processes that saved the team time.
- Transitioned to a contractor role to design and implement scalable solutions using API and AI tools, improving data accuracy, accelerating cross-platform updates to real-time, and managing new Ops Tech interns.

Anne Arundel Community College — Anne Arundel, MD
Volunteer Maker Space Manager — *Aug 2023 - July 2025*

- Maintained and calibrated 3D printers, CNC routers, and a CNC plasma cutter to ensure safe, reliable maker space operations for multiple cohorts.
- Authored a comprehensive maker space guidebook after hands-on testing and research, standardizing usage for new students, reducing troubleshooting for professors, and outlining equipment necessities for staff.

M&T Bank — Wilmington, DE
Software Engineering Intern — *May 2023 - Sep 2023*

- Spearheaded the implementation of Power BI dashboards to improve data visibility for the Data Pirates team, accelerating decision-making for the client.
- Developed and optimized automation tools in C#, .NET, and SQL under Agile & Scrum guidance, increasing process efficiency.

Leadership & Activities & Project Experience

Keep Me Initiative, Duke University — *Nov 2025 - Present*

Co-Founder of Campus-level Material Circularity Initiative, Independent Study, $25k+ grant funding, 2026-2027. Conducting research on best practices for material reuse and recycling, and developing a comprehensive plan for implementation.

Design Climate I & II, Duke University — *Fall 2025 - Spring 2026*

- Identification of need, market, and value proposition for Low-Carbon Cement for Thermal Batteries. Converged into a climate tech solution that implements machine learning to design optimized concrete mixes. Won 3rd place in Duke's 2025 AI hackathon with prototype Climacrete.

MT Lab, University of Delaware — *Aug 2022 - Dec 2022*

- Development of tracking software for microrobots; biomedical applications. Providing the client with control of different parameters and adding an option to change the contrast of the video to aid the tracker in "finding" the blurred microrobot as they move on the z-axis.

Skills

Technical: Python, JavaScript, Typescript, C++, C#, Processing, SQL, Jinja, Firebase, MongoDB, QuickBase/Pipelines, Zapier, RPA, Power BI, CAD/CAM, CNC

Language: Spanish